

Agentic AI in Financial Services

Some Reflections

Manuel Barbero

mbarbero@urcrew.ai

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This document presents an analysis of Agentic AI, its evolution, current capabilities, limitations, and strategic implications for financial services. It explores the transition of AI from descriptive to generative and now to agentic forms, emphasizing the integration of reasoning, planning, and autonomous execution across multiple modalities. It outlines a vision for embedding AI into the knowledge productivity ecosystem, the expected future trajectory of AI, and societal impacts.

EXECUTIVE SUMMARY

Agentic AI promises a significant advancement beyond earlier AI forms by combining perception, reasoning, planning, and autonomous task execution to achieve high-level goals. Unlike traditional descriptive AI — which focuses on statistical methods for classification and prediction — and generative AI, which creates novel content, agentic AI aims to orchestrate complex workflows autonomously across modalities: numbers, text, images, and sound. This evolution expands AI's cognitive capabilities from memorization and classification to generation and strategizing, enabling automation of entire workflows rather than isolated tasks.

Despite its promise, Agentic AI currently faces significant limitations. Its implementation still depends on simplified, rule-based workflows and lacks human-like common sense, intuition, and nuanced contextual understanding. AI systems struggle with non-literal language, ambiguous scenarios, and unforeseen complexities. The scarcity of high-quality training data constrains progress. Larger models risk overfitting and reduced generalizability, while many lack formal mathematical rigor and explainability — critical for trust in high-stakes domains like finance and healthcare.

AI is best deployed as a catalyst for reinventing knowledge work rather than as a siloed technology project. Effective strategy emphasizes stakeholder-centered design, modularity, security, integration, and human-AI collaboration — balancing speed-to-market with scalability, compliance, and human oversight.

AI overall is expected to become abundant, ubiquitous, and cost-effective, lowering barriers for enterprise users to build AI tools. Firms should maintain diversified supplier strategies while developing custom solutions where tight integration with proprietary data and business processes is paramount. Supporting infrastructure for security, usage metering, and quality control remains critical throughout.

A — WHAT IS THE NEW NEWS? GOING BACK BEFORE GOING FORWARD

Cognitive Work and the Evolution of AI Modalities

Cognitive work encompasses the spectrum of mental activities performed by the human brain, mapped along two axes: the modalities (numbers, language, images) and the types of cognitive processes:

- **Memorization:** Storing and recalling information. Humanity has relied on electronics as a force multiplier since the dawn of the digitization age, 70+ years ago.
- **Classification:** Categorizing inputs. AI started here, based on mathematical advances in statistics and linear algebra since the early seventies.
- **Manipulation:** Transforming or reworking information, such as retelling a story or altering an image.
- **Generation:** Creating new artifacts — original narratives, images, or sounds.
- **Reasoning:** Building logical inferences.
- **Planning / Strategizing:** Developing sequences of actions to achieve goals.

The last four cognitive processes have benefited from stupendous advances in AI.

The Evolution of AI: From Descriptive to Generative to Agentic

- **Descriptive (Traditional) AI.** Early systems automated specific, well-defined cognitive tasks in structured data — rule-based logic and supervised learning for classification and prediction. Narrow in scope, requiring human intervention for adaptation.
- **Generative AI.** With advances in deep learning and neural networks, AI evolved to handle unstructured data and generate novel outputs — text, images, audio, code. This marked a significant expansion in modality, moving beyond analysis to creative synthesis.
- **Agentic AI.** The latest leap integrates perception, reasoning, planning, and autonomous execution. Agentic systems are goal-oriented: given a high-level objective, they devise plans, orchestrate tasks, and adapt dynamically across modalities — enabling automation of entire workflows, not just isolated tasks.

Expansion of Modalities

- **Numbers:** Computers have always excelled at numerical analysis. Interestingly, generative AI provides mixed results in mathematics: surprisingly lackluster in actual arithmetic, yet capable in formal proof development and verification [1][2].
- **Language:** NLP models enabled understanding and generation of text.
- **Images & Sound:** Computer vision and audio models brought perception and generation to new sensory domains.
- **Multimodal Integration:** Modern agentic AI can combine modalities, reasoning across text, images, and sound to solve complex problems.

The automation of cognitive work has progressed from descriptive analytics to generative creativity and now to agentic autonomy. Agentic AI represents the convergence of these capabilities, enabling machines to function as intelligent agents that plan, reason, and execute across diverse domains. This evolution is not just technological — it is fundamentally reshaping how organizations leverage AI for productivity, creativity, and strategic decision-making.

B — THE STATE OF AI AND LATEST ADVANCES IN AGENTIC AI: LIMITATIONS

Agentic AI shifts the conversation from *answering questions* to *making things happen*. But there are intrinsic limitations. Despite impressive advancements, AI systems are far from achieving human-like reasoning or autonomous decision-making in complex, real-world contexts.

- **Lack of Common Sense.** AI's reasoning is still largely linear, hampering truly adaptive or creative decisions essential in dynamic environments. AI systems struggle with non-literal language — asked to respond to "Break a leg!", a model might treat it as a literal instruction rather than an idiom conveying good luck.
- **Orchestration still relies on simplified predicate logic.** Multi-agent coordination continues to depend on workflow-based solutions such as decision trees — effective for rule-based tasks but

lacking the flexibility of human judgment. Anything beyond this requires a leap in capabilities that the current state of algorithms, data quality, and computational power cannot yet support.

- **Lack of Quality Data.** The volume of high-quality, comprehensive data needed to train more advanced systems is scarce. Collecting and labeling such data can be expensive, time-consuming, and sometimes impossible.
- **Big is not necessarily Better.** Larger AI models are more prone to overfitting — losing the ability to generalize and adapt to new situations.
- **Lack of Formal Rigor and Explainability.** Many AI systems lack a formal mathematical framework, making it difficult to ensure safety, security, and reliability — or to explain decisions in high-stakes domains like finance, healthcare, or law.
- **LLMs may be a dead-end.** A new type of AI is emerging based on "world models" [3] — systems designed to understand and predict the physical world rather than just process text. A core component is Yann LeCun's **Joint Embedding Predictive Architecture (JEPA)**, which learns by comparing abstract representations of data rather than pixels or words. World models, expected to take up to a decade to fully mature, will be far more suitable for physical applications like robotics and autonomous systems.

In summary: while agentic AI is evolving, its current limitations — reliance on rule-based workflows, linear reasoning, insufficient data quality, and change management hurdles — restrict its ability to perform autonomous, complex decision-making fully.

C — PUTTING AI IN CONTEXT: THE VISION FOR THE KNOWLEDGE PRODUCTIVITY ECOSYSTEM

An effective AI strategy captures today's achievable wins while preparing to absorb tomorrow's breakthroughs — always within guardrails of security, compliance, and human oversight. AI adoption should be framed not as a standalone technology initiative, but as a foundational element in the reinvention of how knowledge workers operate, anchored in a layered architecture that integrates technology, data, user experience, governance, and workflow management.

- **Stakeholder-Centered Design:** Solutions developed with input from content owners, technology teams, and end-users to ensure relevance and usability.
- **Modularity and Reuse:** Build standardized components that can be leveraged across multiple domains and use cases.
- **Security and Governance:** AI embedded within secure, monitored environments with robust access control and policy management serving the entire ecosystem.
- **Integration and Interoperability:** Open APIs and facades abstract complexity, enabling seamless integration with existing systems and workflows.
- **Human-AI Collaboration:** Automation augments — not replaces — human expertise, and knowledge workers can co-create and manage knowledge with AI agents.

Tactically, the overall environment is best understood as a combination of several "market segments," each with specific needs and implementation options — not a centralized, all-encompassing infrastructure: broad productivity tools for general users; highly targeted solutions for domains requiring deep data curation and process integration; and external vendor offerings where capable, configurable solutions exist.

AI initiatives should not be siloed projects, but integrated — or at least loosely coupled — building blocks of a future-ready, knowledge-driven enterprise, aimed at accelerating the emergence of a new paradigm for knowledge work where AI is woven into the fabric of how the organization operates, governs, and innovates.

D — WHERE IS AI LIKELY TO PROGRESS, AND HOW WILL IT IMPACT FINANCIAL SERVICES?

With apologies to Niels Bohr: *"Prediction is hard, especially for the future."* That said, here is an update on reasonably probable planning assumptions:

- **AI will be abundant and ubiquitous** — and therefore probably cheap if not free, like search or the internet. Concerns about skyrocketing AI-cloud expenses may not be justified.
- **A few companies will dominate AI supply** due to immense barriers of entry. A diversified supplier strategy will continue to make sense.
- **The barrier to entry for enterprise users will lower.** Core AI capabilities are becoming a commodity — firms should be ready to build internal solutions knowing they could be made obsolete in the short term.
- **Supporting infrastructure remains an imperative:** consistent security guardrails across LLMs; usage metering; quality controls; and data access — collectively a "Security and Control Plane."
- **Custom AI solutions continue to make sense** when: (i) the use case requires highly relevant, proprietary data, and (ii) tight integration with business process is required. No vendor can profitably build solutions configurable enough for every firm's process idiosyncrasies.

Beyond these structural observations, AI advances will be uneven across domains:

- **"The Verifier Law" [5]:** AI systems perform well when five requisites are met: (i) abundant training data; (ii) ability to run many experiments to evaluate outcomes; (iii) low signal/noise ratio in training data; (iv) domains where humans have already automated decision-making; and (v) ability to unambiguously rate responses as correct or incorrect. Most problems meeting all five criteria have likely already been automated — just not necessarily with AI.
- **Smaller might be better.** Large non-tech industries show inertia; smaller AI-first companies are demonstrating differentiating competitive advantages. While every knowledge worker may eventually gain an AI assistant, this could decrease — not increase — the signal/noise ratio in business communications.
- **"Fast Takeoff" is unlikely** in most domains. In chess, performance has zoomed past human capability since Deep Blue. But in strategic design and architecture, the five Verifier Law requisites are not met — rapid AI dominance is implausible in the near term.
- **"The universe is not infinite, and even if it was, robots don't live in it."** Training AI on surgical procedures would require ethically impossible data collection. Cell-level biological simulation lags immensely behind, say, modeling a chess game. No simulation and no repeated experiment means no training — thus limited AI-powered solutions in such domains.
- **"The Digital Universe is Big — But Nature is Even Bigger" [6]:** Total data created globally in 2023 is estimated at roughly 120 zettabytes; unique, indexed digital knowledge is closer to 10^{14} – 10^{15} bytes. Applying a generous model of human brain knowledge intake, a human over five years processes input in the same order of magnitude as the indexed web. Machines ingest vast quantities quickly but without deep contextual grounding; humans absorb more slowly but integrate knowledge tightly with lived experience.
- **Specialized vs. "General" Intelligence.** Human intelligence is itself highly specialized — an evolutionary toolkit honed for pattern recognition, social reasoning, language, and adaptive problem-solving in uncertain environments. "Artificial General Intelligence" collapses, on inspection, into a collection of domain-specific optimizations. The Verifier's Law offers a predictive lens: AI will keep expanding in domains meeting its criteria; humans will retain superiority in domains with sparse data, ill-structured objectives, and high contextual variability.

E — CONCLUSION(S?)

Humanity's intellectual journey can be imagined as a tiered pyramid — from collecting raw **data** in ancient times, to shaping it into **information** through numerical systems, to recording **knowledge** in books and later digital form. The leap to **insight**, enabled by breakthroughs such as calculus, revealed the laws governing the universe. At the summit remains **wisdom and judgment**, historically the preserve of human minds.

The acceleration brought by digitization and mathematical progress has compressed the timelines for reaching each tier — yet whether AI can fully ascend into the domain of wisdom is still contested. Machines, limited by their disembodied nature, may struggle to capture the non-textual, ephemeral, sensory threads of understanding woven into human decision-making. Bridging this gap through real-time data acquisition might help — but such general, real-time transparency could also create dystopian realities, as narrated in Dave Eggers' *The Circle* [7].

Examples from practice challenge any simple dichotomy. AI can produce an entire campaign structure — themes, slogans, positioning — within minutes. But it was not AI that recognized the strategic necessity for that initiative; that judgment came from lived experience — the kind Bill Duggan describes as *Strategic Intuition* [8]: "knowing that we needed it" was not an AI-generated idea.

Overall, the synthesis of machine speed and human vision is not just a capability — it is a competitive differentiator. Drawing from Elliott Jaques' *Stratified Systems Theory* [9], only a fraction — 10–40% — of high-stratum thinkers will harness AI to deliver work at ten times the speed and quality. The remainder will see much of their contribution commoditized, their economic relevance diminished by factors of two, four, or ten times. This will widen the gulf between the AI-literate and AI-illiterate, a divide with profound market and societal consequences.

As Yuval Noah Harari [10] warns, the displacement of broad swaths of knowledge workers could birth a "Useless Class" — while demand for manual and service roles stabilizes or rises. Whether this triggers social innovations like Universal Basic Income or more troubling outcomes is unknown.

The question is not simply whether AI will reach the summit of human intellect — it is who will be there when it happens. If wisdom and judgment remain human strengths, they will do so because we deliberately choose to cultivate them. Do we worry about our children? Yes — if we neglect AI literacy at scale. No — if we commit to embedding AI fluency in education, career entry, and lifelong skill development. Calls to cease hiring junior talent on the grounds that AI can replicate their work ignore the deeper truth: wisdom and judgment, the apex of the pyramid, cannot simply be engineered — they are learned through practice. If we stop cultivating these skills now, we will have no leaders with the capacity to guide decision-making in a decade's time.

The future is not AI replacing human judgment — it is **strategic co-design of workflows where machines scale and humans decide**. Rather than a binary displacement, the trajectory looks like a mutual expansion of specialized domains. Machines will take over more of the structured, repeatable work; humans will continue to dominate the ambiguous, high-context decisions where verifiability is not possible.

For firms in financial services, this underscores four design imperatives for AI integration:

- Accelerate deployment in **high-verifiability domains** — quantitative analysis, compliance, structured market research.
- **Actively invest in exploration workflows** that might break into low-verifiability territories over time.
- **Re-engineer operational models** to make AI a process engine rather than a decorative tool, accepting that some builds will be short-lived.
- **Maintain a market-wide lens**, anticipating where AI will uplift sectors, displace labor, and trigger regulatory or societal shifts that could reshape investment landscapes.



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